Graduate Level Elementary Particle Physics

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*https://github.com/yuriphestos/gradphestos-epp

3 Task 3

3.1 Pion Decay Constant f_π

Calculate the probability (i.e. the partial width and branching ratio) of exclusive decay $\tau \to \nu_\tau \pi^-$ in terms of the parameter f_π. Using the comparison with PDG, determine f_π, and compare its value with the one that can be extracted from $\pi \to \mu\nu$ decay rate.

$$\tau(p_1) \quad \to \quad \nu_\tau(p_2)\pi^-(p_3)$$

The effective Lagrangian for this weak charged-current transition is:

$$\mathcal{L}_{eff} = \frac{G_F \cos\theta_c}{\sqrt{2}} \left[\bar{u}_\nu \gamma_\mu (1-\gamma_5) u_\tau\right] \left[\bar{u}_d \gamma^\mu (1-\gamma_5) u_u\right]$$

To evaluate the hadronic matrix element, we parameterize the vacuum-to-pion transition of the axial-vector current using the pion decay constant f_π. Because the pion is in the final state, it is created from the vacuum:

$$\langle \pi^-(p_3)|\bar{u}_d \gamma^\mu (1-\gamma_5) u_u|0\rangle = i f_\pi p_3^\mu$$

Thus, the matrix element for the decay becomes:

$$\mathcal{M} = \frac{G_F \cos\theta_c}{\sqrt{2}} (i f_\pi p_3^\mu) \left[\bar{u}_\nu \gamma_\mu (1-\gamma_5) u_\tau\right]$$

Using momentum conservation ($p_3 = p_1 - p_2$) and the Dirac equation for the on-shell leptons ($\bar{u}_\nu \not{p}_2 = 0$ and $\not{p}_1 u_\tau = m_\tau u_\tau$), we can contract the momentum with the leptonic current:

$$p_3^\mu \left[\bar{u}_\nu \gamma_\mu (1-\gamma_5) u_\tau\right] = \bar{u}_\nu (\not{p}_1 - \not{p}_2)(1-\gamma_5) u_\tau = m_\tau \left[\bar{u}_\nu (1+\gamma_5) u_\tau\right]$$

Yielding the simplified form:

$$\mathcal{M} = \frac{G_F \cos\theta_c}{\sqrt{2}} (i f_\pi m_\tau) \left[\bar{u}_\nu (1+\gamma_5) u_\tau\right]$$

Squaring the amplitude and averaging over the initial τ spin:

$$\langle |\mathcal{M}|^2 \rangle = \left(\frac{1}{2}\right) \frac{G_F^2 \cos^2\theta_c}{2} f_\pi^2 m_\tau^2 (8 p_1 \cdot p_2) = G_F^2 \cos^2\theta_c f_\pi^2 m_\tau^4 \left(1 - \frac{m_\pi^2}{m_\tau^2}\right)$$

where we utilized the kinematic relation $2(p_1 \cdot p_2) = m_\tau^2 - m_\pi^2$. The 2-body decay rate is (PDG 49.18):

$$\Gamma(\tau \to \nu_\tau \pi^-) = \frac{|\mathcal{M}|^2}{8\pi} \frac{|\vec{p}_3|}{m_\tau^2} = \frac{|\mathcal{M}|^2}{8\pi} \frac{m_\tau^2}{2m_\tau^3} \left(1 - \frac{m_\pi^2}{m_\tau^2}\right) = \boxed{\frac{G_F^2 \cos^2\theta_c}{16\pi} f_\pi^2 m_\tau^3 \left(1 - \frac{m_\pi^2}{m_\tau^2}\right)^2}$$
$$= (1.42 \times 10^{-11}) f_\pi^2 \ \text{GeV}^{-1}$$

From the PDG, the branching ratio of $\tau \to \nu_\tau \pi^-$ is 10.82 % and the mean lifetime of τ is 290.3×10^{-15} s. The experimental partial width is therefore:

$$\Gamma_{exp}(\tau \to \nu_\tau \pi^-) = \frac{6.58 \times 10^{-25} \ \text{GeV}}{290.3 \times 10^{-15}} \times 0.1082 = 2.45 \times 10^{-13} \ \text{GeV}$$

Equating the theoretical and experimental widths, the pion decay constant f_π is:

$$f_\pi = \sqrt{\frac{2.45 \times 10^{-13}}{1.42 \times 10^{-11}}}\ \text{GeV} = 131.4\ \text{MeV} \tag{3.1}$$

Comparison with $\pi^+ \to \mu^+\nu_\mu$:

By crossing symmetry, the decay rate for $\pi^+ \to \mu^+\nu_\mu$ relies on the identical effective vertex, yielding the well-known width:

$$\Gamma(\pi^+ \to \mu^+\nu_\mu) = \frac{G_F^2 \cos^2\theta_c}{8\pi} f_\pi^2 m_\mu^2 m_\pi \left(1 - \frac{m_\mu^2}{m_\pi^2}\right)^2 = (1.46 \times 10^{-15}) f_\pi^2\ \text{GeV}^{-1}$$

Using the PDG values for the pion mean lifetime (2.6×10^{-8} s) and branching ratio (99.988 %):

$$f_\pi = \sqrt{\frac{6.58 \times 10^{-25}}{2.6 \times 10^{-8}} \times 0.99988 \times \frac{1}{1.46 \times 10^{-15}}}\ \text{GeV} = 131.6\ \text{MeV} \tag{3.2}$$

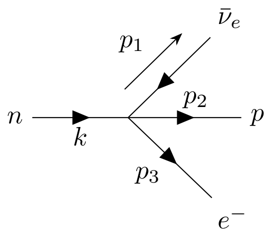
This value extracted from pion decay is in excellent agreement with the value of (3.1) extracted from tau decay, demonstrating the universality of the weak interaction and the axial-vector current.

The strict isospin symmetry limit assumes identical up and down quark masses and neglects electromagnetic corrections, implying $f_{\pi^\pm} = f_{\pi^0}$. However, explicit isospin breaking ($m_d > m_u$ and $q_u \neq q_d$) causes the physical values to differ slightly: $f_{\pi^\pm} \approx 130.4$ MeV while the neutral pion decay constant is $f_{\pi^0} \approx 130.0$ MeV.

3.2 Polarized Neutron Decay

Calculate the differential decay rate for the polarized neutrons wrt to the angle of the final state electron. Obtain the expression for angular asymmetry in terms of the g_A coupling constant (Use $m_e/m_n \to 0$ limit). Using the results of the PERKEO experiment, and the formula for the neutron lifetime, make a prediction for the τ_n. Useful refs: `https://arxiv.org/pdf/1812.04666.pdf` **and** `https://arxiv.org/pdf/1802.01804.pdf`**.**

Neutron β-decay



Interaction and differential decay rate (PDG 49.20):

$$\mathcal{L} = \frac{G_F \cos\theta_c}{\sqrt{2}} \left[\bar{u}_p \gamma_\mu (1 - g_A \gamma_5) u_n\right] \left[\bar{u}_e \gamma^\mu (1 - \gamma_5) v_\nu\right]$$

Note that the vector coupling $g_V = 1$ due to CVC. The differential decay rate is:

$$d\Gamma = \frac{1}{(2\pi)^5} \frac{1}{2m_n} \frac{1}{2E_p} |\mathcal{M}|^2 |p_1^*||p_3| dm_{12} d\Omega_1^* d\Omega_3 = \frac{1}{(2\pi)^5} \frac{1}{4m_n^2} |\mathcal{M}|^2 E_\nu |\vec{p}_e| dE_e d\Omega_e d\Omega_\nu$$

where $E_\nu \simeq m_n - m_p - E_e$. Using the Dirac spinors with the non-relativistic assumptions $E \simeq m$ and $p \ll E$ for the neutron and proton:

$$\psi = \sqrt{E+m} \begin{pmatrix} \chi \\ \frac{\vec{\sigma}\cdot\vec{p}}{E+m}\chi \end{pmatrix} \simeq \sqrt{2m} \begin{pmatrix} \chi \\ 0 \end{pmatrix}$$

This simplifies the hadronic bilinear covariants:

$$\bar{\psi}_p \gamma_0 \psi_n = \psi_p^\dagger \psi_n = (2m_n) \chi_p^\dagger \chi_n \qquad \text{(Spatial components } \bar{\psi}_p \gamma_i \psi_n \approx 0\text{)}$$
$$\bar{\psi}_p \gamma_i \gamma_5 \psi_n = \psi_p^\dagger \sigma_i \psi_n = (2m_n) \chi_p^\dagger \sigma_i \chi_n \qquad \text{(Time component } \bar{\psi}_p \gamma_0 \gamma_5 \psi_n \approx 0\text{)}$$

Squaring the amplitude $|\mathcal{M}|^2 = (G_F^2 \cos^2\theta_c/2) T^h_{\mu\nu} T_l^{\mu\nu}$, we construct the hadronic tensor for a polarized neutron ($\sum_{\text{spins}} \chi_p \chi_p^\dagger = 1$):

$$\begin{aligned}
T^h_{00} &= (2m_n)^2 \chi_n^\dagger \chi_p \chi_p^\dagger \chi_n = (2m_n)^2 \\
T^h_{0i} &= T^h_{i0} = -g_A (2m_n)^2 \chi_n^\dagger \sigma_i \chi_n \\
T^h_{ij} &= g_A^2 (2m_n)^2 \chi_n^\dagger \sigma_i \sigma_j \chi_n = g_A^2 (2m_n)^2 \chi_n^\dagger (\delta_{ij} + i\varepsilon_{ijk}\sigma_k) \chi_n
\end{aligned}$$

The leptonic tensor is standard:

$$T_l^{\mu\nu} = \mathrm{Tr}\left[\gamma^\mu (1-\gamma_5) \not{p}_\nu \gamma^\nu (1-\gamma_5) \not{p}_e\right] = 8\left[p_\nu^\mu p_e^\nu + p_e^\mu p_\nu^\nu - g^{\mu\nu}(p_\nu \cdot p_e) + i\varepsilon^{\mu\nu\alpha\beta} p_{\nu,\alpha} p_{e,\beta}\right]$$

Contracting the tensors and integrating over the unobserved neutrino solid angle ($\int d\Omega_\nu$) eliminates terms proportional to $\vec{p}_\nu$. For a neutron polarized along $\pm\hat{z}$, $\chi_n^\dagger \vec{\sigma} \chi_n = \hat{s} = \pm\hat{z}$. Keeping only terms dependent on the electron direction $\hat{p}_e$:

$$T^h_{\mu\nu}T^{\mu\nu}_l = 32m_n^2 E_\nu \left[(1+3g_A^2)E_e - 2|\vec{p}_e|g_A(g_A-1)\cos\theta_e\right]$$

The differential decay rate becomes:

$$d\Gamma = \frac{G_F^2\cos^2\theta_c}{8\pi^4}E_\nu^2|\vec{p}_e|\left[(1+3g_A^2)E_e - 2|\vec{p}_e|g_A(g_A-1)\cos\theta_e\right]dE_e d\Omega_e$$

To find the angular asymmetry A, we apply the massless electron limit ($m_e/m_n \to 0 \implies |\vec{p}_e| \approx E_e$). The maximum electron energy is dictated by kinematics: $E_{\max} = m_n - m_p$. Therefore, $E_\nu = E_{\max} - E_e$. Integrating out the electron energy:

$$\frac{d\Gamma}{d\Omega_e} \propto \int_0^{E_{\max}} (E_{\max}-E_e)^2 E_e^2 \left[(1+3g_A^2) - 2g_A(g_A-1)\cos\theta_e\right]dE_e$$

The polynomial integral evaluates smoothly:

$$\int_0^{E_{\max}} \left(E_{\max}^2 E_e^2 - 2E_{\max}E_e^3 + E_e^4\right)dE_e = \frac{E_{\max}^5}{30}$$

Factoring out the $(1+3g_A^2)$ term from the remaining bracket gives the decay distribution in the standard form $(1+A\cos\theta_e)$:

$$\frac{d\Gamma}{d\Omega_e} \propto \frac{E_{\max}^5}{30}(1+3g_A^2)\left[1 - 2\frac{g_A(g_A-1)}{1+3g_A^2}\cos\theta_e\right]$$

This yields the standard expression for the angular asymmetry:

$$\boxed{A = -2\frac{g_A(g_A-1)}{1+3g_A^2}} \tag{3.3}$$

<u>Comparison with PERKEO and Neutron Lifetime</u>:

The PERKEO experiment (1812.04666 [nucl-ex]) precisely measured the asymmetry to be:

$$A = -0.11985$$

Equating this with (3.3) and solving the quadratic equation for g_A yields the coupling constant:

$$\boxed{g_A \simeq 1.276}$$

Using this extracted g_A and the formula for the neutron lifetime from 1802.01804 [hep-ph] ($\tau_n \propto (1+3g_A^2)^{-1}$):

$$\tau_n(1+3g_A^2) = 5172 \text{ s} \quad \Rightarrow \quad \boxed{\tau_n = \frac{5172}{1+3(1.276)^2} \simeq 878.6 \text{ s}}$$

This prediction for the neutron lifetime is in excellent agreement with the physical value of ~ 879.4 s cited in the paper, demonstrating the remarkable consistency of the Standard Model weak interaction parameters.

3.3 Solar Neutrino-Electron Scattering

Neutrino scattering on electrons (which we can calculate if we want to) is given by neutral currents for ν_μ and ν_τ, and by charged and neutral currents for ν_e. The corresponding cross sections can be found *e.g.* in Ref. Phys.Rev.D 63 (2001) 112001 e-Print: hep-ex/0101039 [hep-ex]. Compare the size of the ν_e cross section with that of ν_μ or ν_τ. Use the predicted value for the Boron-8 neutrino flux from the Sun to estimate the event rate per ton of liquid scintillator, assuming that all neutrinos arrive as ν_e. Using more detailed distribution of Boron-8 neutrino over energy (that you can find at `http://www.sns.ias.edu/jnb/` and references therein), compare the predicted event rate and observed event rate by the Borexino collaboration (Take $E_e > 5\,$MeV), `https://arxiv.org/pdf/0808.2868.pdf`, Fig 6 and 7.

<u>Cross Section Comparison</u>:

From 0101039 [hep-ex], the total cross sections for elastic ν–e scattering are:

$$\sigma^{\nu_\mu} = \sigma_0 \left(g_L^2 + \frac{g_R^2}{3} \right) ; \qquad \sigma^{\nu_e} = \sigma_0 \left[(g_L + 2)^2 + \frac{g_R^2}{3} \right]$$

where $\sigma_0 = 2G_F^2 m_e E_\nu / \pi$. Using $\sin^2\theta_W = 0.231$, the weak couplings are:

$$g_L = 2\sin^2\theta_W - 1 \approx -0.538 \,, \qquad g_R = 2\sin^2\theta_W \approx 0.462$$

For ν_e, the shift $g_L \to g_L + 2$ arises from the charged current W-exchange amplitude, which interferes constructively with the neutral current Z-exchange. The ratio of total cross sections is:

$$\frac{\sigma^{\nu_\mu}}{\sigma^{\nu_e}} = \frac{g_L^2 + g_R^2/3}{(g_L+2)^2 + g_R^2/3} = \frac{(-0.538)^2 + (0.462)^2/3}{(1.462)^2 + (0.462)^2/3} = \frac{0.360}{2.209} \simeq 0.163$$

The $\nu_e - e$ cross section is roughly 6 times larger than $\nu_\mu - e$, due to the coherent CC+NC contribution.

<u>Event Rate Estimation</u>:

The expected rate is given by integrating over the incoming neutrino energy and the electron recoil energy:

$$R = N_e \int_{E_{e,\min}}^{E_{e,\max}} dE_e \int_{E_\nu^{\min}(E_e)}^{E_\nu^{\max}} dE_\nu \, \frac{d\sigma^{\nu_e}}{dE_e} \cdot \frac{d\Phi}{dE_\nu}$$

where the differential cross section with respect to the electron recoil energy is obtained from $y = E_e/E_\nu$ via the Jacobian $dy = dE_e/E_\nu$:

$$\frac{d\sigma^{\nu_e}}{dE_e} = \frac{1}{E_\nu}\frac{d\sigma^{\nu_e}}{dy} = \frac{G_F^2 m_e}{2\pi} \left[(g_L+2)^2 + g_R^2 \left(1 - \frac{E_e}{E_\nu} \right)^2 \right]$$

Note that the prefactor $G_F^2 m_e/(2\pi)$ carries no explicit E_ν dependence; the neutrino energy enters only through the $(1 - E_e/E_\nu)^2$ term and through the kinematic limits. The maximum electron recoil energy for a given E_ν is:

$$E_e^{\max}(E_\nu) = \frac{2E_\nu^2}{m_e + 2E_\nu}$$

Inverting this relation gives the minimum neutrino energy required to produce a recoil E_e:

$$E_\nu^{\min}(E_e) = \frac{1}{2}\left(E_e + \sqrt{E_e^2 + 2m_e E_e}\right) \approx E_e + \frac{m_e}{4}$$

where the approximation holds for $E_e \gg m_e$. For the Borexino analysis threshold $E_e > 5$ MeV, this gives $E_\nu^{\min} \simeq 5.25$ MeV.

The number of target electrons N_e in 100 tons of Borexino liquid scintillator (pseudocumene, C_9H_{12}, with 66 electrons per molecule, $M = 120.19\,\text{g/mol}$) is:

$$N_e = \frac{10^8 \text{ g}}{120.19 \text{ g/mol}} \times N_A \times 66 \simeq 3.3 \times 10^{31} \text{ electrons per 100 ton}$$

Using the ^{8}B solar neutrino flux spectrum from Bahcall's tables (BS05 SSM), normalized to the predicted total flux $\Phi_{\text{total}} = 5.69 \times 10^6 \text{ cm}^{-2}\text{s}^{-1}$, and integrating numerically over the kinematic phase space, the expected rate assuming all neutrinos arrive as ν_e (no oscillation hypothesis) is:

$$\boxed{R_{\text{predicted}} \simeq 0.33 \text{ cpd/100 t}} \qquad (\text{for } E_e > 5 \text{ MeV})$$

Comparison with Borexino and Evidence for Neutrino Oscillations:

The Borexino collaboration (0808.2868 [astro-ph], Figs. 6 and 7) reports:

$$R_{\text{observed}} \simeq 0.13 \text{ cpd/100 t} \qquad (\text{for } E_e > 5 \text{ MeV})$$

The predicted no-oscillation rate exceeds the observed rate by more than a factor of two. This suppression is the direct signature of solar neutrino flavor oscillations. Because a large fraction of the solar ν_e flux undergoes MSW resonant conversion inside the Sun, the neutrinos arrive at Earth as a mixture of flavors. The observed rate can be written as:

$$R_{\text{observed}} = R_{\text{predicted}} \left[P_{ee} + (1 - P_{ee})\,\frac{\sigma^{\nu_\mu}}{\sigma^{\nu_e}}\right]$$

where P_{ee} is the ν_e survival probability. Using $\sigma^{\nu_\mu}/\sigma^{\nu_e} \simeq 0.163$:

$$\frac{R_{\text{observed}}}{R_{\text{predicted}}} = \frac{0.13}{0.33} \simeq 0.40 = P_{ee} + 0.163\,(1 - P_{ee}) = 0.837\,P_{ee} + 0.163$$

Solving for the survival probability:

$$\boxed{P_{ee} \simeq \frac{0.40 - 0.163}{0.837} \simeq 0.28}$$

This is in excellent agreement with the MSW-LMA prediction of $P_{ee} \approx 0.3 - 0.35$ for ^{8}B neutrino energies $(5 - 15$ MeV), providing direct evidence for solar neutrino oscillations in the elastic scattering channel.

(see `task-3-3.py`)

3.4 Constraining New Physics with Pion Decay

Imagine there is some "new physics" operator in the Lagrangian given by

$$\Delta\mathcal{L}_{NP} = \frac{1}{(\Lambda_{NP})^2}[\bar{\nu}_e(1+\gamma_5)e][\bar{d}(1-\gamma_5)u]$$

Using the data from the latest experiment, PiENu collaboration, Phys.Rev.Lett. 115 (2015) no.7, 071801, (`https://arxiv.org/pdf/1506.05845.pdf`), and assuming perfect agreement with the Standard Model, set the constraint on the size of Λ_{NP}. You will need a matrix element, $\langle 0|\bar{d}\gamma_5 u|\pi\rangle$. I give it to you as: $\langle 0|\bar{d}\gamma_5 u|\pi\rangle = f_\pi \times (m_u+m_d)^{-1}m_\pi^2$, but you may want to derive it yourself using the definition of f_π.

The given "New Physics" Lagrangian for $\pi^+ \to e^+\nu_e$ is

$$\Delta\mathcal{L}_{NP} = \frac{1}{(\Lambda_{NP})^2}[\bar{\nu}_e(1+\gamma_5)e][\bar{d}(1-\gamma_5)u]$$

To evaluate the hadronic matrix element, we start with the standard definition of the pion decay constant via the axial-vector current:

$$\langle 0|\bar{d}\gamma^\mu\gamma_5 u|\pi^+(p)\rangle = if_\pi p^\mu$$

Taking the 4-divergence (∂_μ) of this matrix element pulls down a factor of $-ip_\mu$:

$$\langle 0|\partial_\mu(\bar{d}\gamma^\mu\gamma_5 u)|\pi^+(p)\rangle = -ip_\mu(if_\pi p^\mu) = f_\pi m_\pi^2$$

Using the Dirac equation, the divergence of the axial current is $\partial_\mu(\bar{d}\gamma^\mu\gamma_5 u) = i(m_u+m_d)\bar{d}\gamma_5 u$ (the relative plus sign between the quark masses arises from $\{\gamma_5,\gamma^\mu\}=0$ when applying the equations of motion to both quark fields). Equating the two yields the required scalar matrix element:

$$i(m_u+m_d)\langle 0|\bar{d}\gamma_5 u|\pi^+\rangle = f_\pi m_\pi^2 \quad \Rightarrow \quad \boxed{\langle 0|\bar{d}\gamma_5 u|\pi^+\rangle = -i\frac{f_\pi m_\pi^2}{m_u+m_d}}$$

Because the pion is a pseudoscalar, the vector component vanishes by parity ($\langle 0|\bar{d}u|\pi^+\rangle = 0$), leaving only the pseudoscalar term for the New Physics hadronic transition:

$$\langle 0|\bar{d}(1-\gamma_5)u|\pi^+\rangle = -\langle 0|\bar{d}\gamma_5 u|\pi^+\rangle = i\frac{f_\pi m_\pi^2}{m_u+m_d}$$

The New Physics amplitude is therefore:

$$\boxed{\mathcal{M}_{NP} = \frac{if_\pi}{(\Lambda_{NP})^2}\frac{m_\pi^2}{m_u+m_d}[\bar{\nu}_e(1+\gamma_5)e]}$$

Squaring this amplitude and utilizing $2(p_e\cdot p_\nu) = m_\pi^2 - m_e^2$:

$$|\mathcal{M}_{NP}|^2 = \frac{f_\pi^2}{(\Lambda_{NP})^4}\left(\frac{m_\pi^2}{m_u+m_d}\right)^2(8p_e\cdot p_\nu) = \frac{4f_\pi^2 m_\pi^2}{(\Lambda_{NP})^4}\left(\frac{m_\pi^2}{m_u+m_d}\right)^2\left(1-\frac{m_e^2}{m_\pi^2}\right)$$

For comparison, the Standard Model amplitude is evaluated via the $V-A$ current. Contracting the pion momentum with the leptonic current via the Dirac equation (as derived in Section 3.1), and noting that $\not{p}_e\, v_e =$

$-m_e\, v_e$ for the positron spinor:

$$\boxed{\mathcal{M}_{SM} = -i\frac{G_F\cos\theta_c}{\sqrt{2}} f_\pi m_e[\bar{\nu}_e(1+\gamma_5)e]}$$

The SM amplitude is proportional to the electron mass m_e due to helicity suppression. Because the pion is spin-0 and the decay products are back-to-back, angular momentum conservation dictates their spins must be parallel. Since the SM strictly creates left-handed neutrinos, the positron is forced into a left-handed helicity state. However, the $V-A$ interaction only couples to right-handed antifermions. The necessary chirality-flip is proportional to m_e/E, heavily suppressing the decay.

Because both amplitudes carry identical spinor structures $[\bar{\nu}_e(1+\gamma_5)e]$, they interfere perfectly. The total amplitude is:

$$\mathcal{M} = \mathcal{M}_{SM} + \mathcal{M}_{NP} = \mathcal{M}_{SM}\left[1 - \frac{\sqrt{2}}{G_F\cos\theta_c}\frac{1}{m_e}\frac{1}{(\Lambda_{NP})^2}\frac{m_\pi^2}{m_u+m_d} + \cdots\right] \equiv \mathcal{M}_{SM}\left[1+\delta_{NP}\right]$$

where the minus sign reflects the relative sign between the SM and NP amplitudes (destructive interference). Squaring the total amplitude yields $|\mathcal{M}|^2 \approx |\mathcal{M}_{SM}|^2(1+2\delta_{NP})$ to leading order in the new physics scale, where $\delta_{NP} \propto \Lambda_{NP}^{-2}$. Thus, the total decay width is:

$$\Gamma_{\pi\to e\nu} = \Gamma_{SM}(1+2\delta_{NP})$$

Since the proposed operator strictly alters the electron channel, the muon decay rate $\Gamma_{\pi\to\mu\nu}$ is unaffected. The theoretical branching ratio becomes:

$$R_{e/\mu}^{NP} = \frac{\Gamma_{\pi\to e\nu}}{\Gamma_{\pi\to\mu\nu}} = \underbrace{\frac{\Gamma_{SM,e}}{\Gamma_{SM,\mu}}}_{R_{e/\mu}^{SM}} + \underbrace{R_{e/\mu}^{SM}\cdot 2\delta_{NP}}_{\Delta R_{e/\mu}^{NP}}$$

The 2015 PiENu collaboration (1506.05845 [hep-ex]) reports the highly precise experimental ratio alongside the theoretical Standard Model prediction:

$$\begin{aligned} R_{e/\mu}^{Exp} &= (1.2344 \pm 0.0023_{\text{stat}} \pm 0.0019_{\text{syst}}) \times 10^{-4} \\ R_{e/\mu}^{SM} &= (1.2352 \pm 0.0001) \times 10^{-4} \end{aligned}$$

Assuming perfect agreement with the Standard Model, the allowed window for New Physics is constrained by the total experimental uncertainty $\delta R = \sqrt{0.0023^2 + 0.0019^2} \times 10^{-4} \approx 0.0030 \times 10^{-4}$. We therefore demand $|\Delta R_{e/\mu}^{NP}| \leq 0.0030 \times 10^{-4}$.

Writing $\Delta R_{e/\mu}^{NP} = 2|\delta_{NP}| \times R_{e/\mu}^{SM}$ explicitly:

$$R_{e/\mu}^{SM}\cdot\frac{2\sqrt{2}}{G_F\cos\theta_c}\cdot\frac{1}{m_e}\cdot\frac{m_\pi^2}{m_u+m_d}\cdot\frac{1}{(\Lambda_{NP})^2} \leq 0.0030\times 10^{-4}$$

Using $\overline{\text{MS}}$ quark masses at $\mu = 2$ GeV from the PDG ($m_u \simeq 2.16$ MeV, $m_d \simeq 4.67$ MeV, so $m_u+m_d \simeq 6.8$ MeV):

$$\Lambda_{NP}^2 \geq \frac{R_{e/\mu}^{SM}}{0.0030\times 10^{-4}}\cdot\frac{2\sqrt{2}}{G_F\cos\theta_c}\cdot\frac{1}{m_e}\cdot\frac{m_\pi^2}{m_u+m_d}$$

Evaluating each factor:

$$\frac{R_{e/\mu}^{SM}}{\delta R} = \frac{1.2352}{0.0030} \simeq 412\,, \quad \frac{2\sqrt{2}}{G_F \cos\theta_c} \simeq 2.48 \times 10^5 \text{ GeV}^2\,, \quad \frac{1}{m_e} \simeq 1955 \text{ GeV}^{-1}\,, \quad \frac{m_\pi^2}{m_u + m_d} \simeq 2.87 \text{ GeV}$$

$$\Lambda_{NP}^2 \geq 412 \times 2.48 \times 10^5 \times 1955 \times 2.87 \simeq 5.7 \times 10^{11} \text{ GeV}^2$$

Therefore, to remain hidden within the experimental error bars of the PiENu data, this specific New Physics operator is constrained to energy scales:

$$\boxed{\Lambda_{NP} \gtrsim 7.5 \times 10^5 \text{ GeV} \sim 750 \text{ TeV}}$$